

Ordinary Differential Equations

ASSIGNMENT 1 – MA 417-2025

TOPIC : METHOD OF SOLUTIONS AND PRELIMINARIES.
 05.08.2025

1.1 Let $y: [a, b] \rightarrow \mathbb{R}$ be a continuous function which satisfies the following relation.

$$\frac{1}{2}y^2(t) \leq \frac{1}{2}y_0^2 + \int_a^t \Psi(s)y(s) \, ds, \quad \forall t \in [a, b].$$

Where $a, b, y_0 \in \mathbb{R}$ and Ψ is a nonnegative continuous function on $[a, b]$. Show that

$$|y(t)| \leq |y_0| + \int_a^t \Psi(s) \, ds.$$

Solution. For a fixed $\epsilon > 0$, define $w_\epsilon(t) = y_0^2 + \epsilon^2 + 2 \int_a^t \Psi(s)y(s) \, ds$. Notice $w_\epsilon(t) > 0$ for all $a \leq t \leq b$. Then it is easy to see that $w'_\epsilon(t) = 2\Psi(t)y(t) \leq 2\Psi(t)\sqrt{w_\epsilon(t)}$ for all $t \in (a, b)$. This implies upon integrating on the interval $[a, t]$,

$$|y(t)| < \sqrt{w_\epsilon(t)} \leq \sqrt{w_\epsilon(a)} + \int_a^t \Psi(s) \, ds = \sqrt{y_0^2 + \epsilon^2} + \int_a^t \Psi(s) \, ds.$$

Now letting $\epsilon \rightarrow 0^+$, we obtain the desired inequality.

1.2 Suppose $x(t), \Phi(t)$ are continuous functions that satisfy

$$x(t) \leq x(\tau) + \int_\tau^t \Phi(s)x(s) \, ds$$

for all $t, \tau \in (a, b)$. Show that for any fixed $t_0 \in (a, b)$,

$$x(t) = x(t_0)e^{\int_{t_0}^t \Phi(s) \, ds}.$$

Solution. Fix $t_0 \in (a, b)$. Notice $x(t) \leq x(t_0) + \int_{t_0}^t \Phi(s)x(s) \, ds$ also $x(t_0) \leq x(t) - \int_{t_0}^t \Phi(s)x(s) \, ds$. Thus $x(t) = x(t_0) + \int_{t_0}^t \Phi(s)x(s) \, ds$, which implies $x'(s) = \Phi(s)x(s)$ for all $s \in (a, b)$. Therefore, using the integrating factor $\exp \int_s^{t_0} \Phi(r) \, dr$, one has $\frac{d}{ds}[x(s) \exp \int_s^{t_0} \Phi(r) \, dr] = 0$ and this concludes the result.

1.3 The equation

$$y'(t) + P(t)y(t) = Q(t)y^n(t)$$

which is known as the *Bernoulli's equation*, is linear when $n = 0, 1$. Show that it can be reduced to a linear equation for any other value of n by the change of variable $z = y^{1-n}$ and apply this method to solve the following equation:

$$ty^2(t)y'(t) + y^3(t) = t \cos t, \quad t \in \mathbb{R}.$$

1.4 If the Wronskian W of the functions f and g satisfies, $W(f, g)(t) = 3e^{4t}$, and $f(t) = e^{2t}$ for all $t \in \mathbb{R}$, find $g(t)$.

Solution.

$$\boxed{g(t) = (3t + C)e^{2t}, \quad C \in \mathbb{R}.$$

1.5 Find the general solutions of each of the following equations.

$$(a) \quad y'(t) = \frac{t}{1+t}.$$

$$(b) \quad y'(t) = \frac{t^2+2}{y(t)}.$$

1.6 Use variation of parameters method to find the general solution of the following equation.

$$y''(t) - y(t) = \sin y.$$

1.7 Find a necessary and sufficient condition on the constants $a, b \in \mathbb{R}$ such that the ODE

$$u''(t) + au'(t) + bu(t) = 0, \quad t \in \mathbb{R}$$

admits a bounded, nontrivial (i.e. $u(t) \neq 0$ for some $t \in \mathbb{R}$) solution.

Solution. The characteristic equation for the given problem is

$$x^2 + ax + b = 0.$$

Therefore solutions of the characteristic equation are $\frac{-a \pm \sqrt{a^2 - 4b}}{2}$. General solution of the given ODE are $c_1 e^{\left(\frac{-a + \sqrt{a^2 - 4b}}{2}\right)t} + c_2 e^{\left(\frac{-a - \sqrt{a^2 - 4b}}{2}\right)t}$ for $t \in \mathbb{R}$, provided characteristic equation has different roots, otherwise general solutions are $c_1 e^{\frac{a}{2}t} + c_2 t e^{\frac{a}{2}t}$, where $c_1, c_2 \in \mathbb{R}$. Clearly, for real roots of characteristic equation, we have unbounded solution. So roots must be complex number. Suppose $p + iq$, $p, q \in \mathbb{R}$ is a root of the characteristic equation, (hence, other root is $p - iq$). Then $e^{pt} \cos(qt)$ and $e^{pt} \sin(qt)$ forms a basis for the vector space of solution set. Hence for bounded nontrivial solution, roots must be purely imaginary number. Therefore a necessary and sufficient condition for bounded, nontrivial solution is $a = 0$ and $a^2 - 4b < 0$. Equivalently, a necessary and sufficient condition for bounded, nontrivial solution is $a = 0$ and $b > 0$.

1.8 Define for $t \in \mathbb{R}$,

$$\begin{aligned} u_{i,j}(t) &= t^j e^{\lambda_i t}, & 1 \leq i \leq m, 0 \leq j \leq m_i. \\ v_{k,l}(t) &= t^l e^{x_k t} \sin(y_k t), & 1 \leq k \leq n, 0 \leq l \leq n_k. \\ w_{k,l}(t) &= t^l e^{x_k t} \cos(y_k t), & 1 \leq k \leq n, 0 \leq l \leq n_k. \end{aligned}$$

where m, n, m_i, n_k are non-negative integers and λ_i , $i = 1, \dots, m$; x_k, y_k , $k = 1, \dots, n$ are given real numbers. Show that the functions $\{u_{i,j}, v_{k,l}, w_{k,l}\}$ where i, j, k, l vary as above are linearly independent.

Solution. We will use a basic linear algebra fact that if A and B are two set of linearly independent vectors of a vector space such that $\text{span}\{A\} \cap \text{span}\{B\} = \{0\}$, then $A \cup B$ is also linearly independent. Now we claim that if i is fixed and $\lambda_1, \dots, \lambda_m$ are distincts then $\{t^i e^{\lambda_1 t}, \dots, t^i e^{\lambda_m t}\}$ is linearly independent set. Indeed, consider the relation

$$\begin{aligned} c_1 t^i e^{\lambda_1 t} + \dots + c_m t^i e^{\lambda_m t} &= 0 \\ \Rightarrow t^i (c_1 e^{\lambda_1 t} + \dots + c_m e^{\lambda_m t}) &= 0 \\ \Rightarrow c_1 e^{\lambda_1 t} + \dots + c_m e^{\lambda_m t} &= 0. \end{aligned} \tag{1}$$

One can show that $\{e^{\lambda_1 t}, \dots, e^{\lambda_m t}\}$ is linearly independent set. To show this, differentiate (1) with respect to t , $(m-1)$ times and then solve the system, which yields $c_1 = \dots = c_m = 0$. Then $\{t^i e^{\lambda_1 t}, \dots, t^i e^{\lambda_m t}\}$ is linearly independent set for fixed i . Now using the linear algebra fact that is stated above, we get that the family of functions $\{u_{i,j}(t)\}$, where $1 \leq i \leq m, 0 \leq j \leq m_i$ is linearly independent.

Now to prove that $\{u_{i,j}(t), v_{k,l}(t), w_{k,l}(t)\}$ are linearly independent, for real $a_{i,j}, b_{k,l}, c_{k,l}$ consider the relation

$$\begin{aligned} \sum_{i=1}^m \sum_{j=0}^{m_i} a_{i,j} t^j e^{\lambda_i t} + \sum_{k=1}^n \sum_{l=0}^{n_k} b_{k,l} t^l e^{x_k t} \sin(y_k t) + \sum_{k=1}^n \sum_{l=0}^{n_k} c_{k,l} t^l e^{x_k t} \cos(y_k t) &= 0 \\ \Rightarrow \sum_{i=1}^m \sum_{j=0}^{m_i} a_{i,j} t^j e^{\lambda_i t} + \sum_{k=1}^n \sum_{l=0}^{n_k} (c_{k,l} - i b_{k,l}) t^l e^{(x_k + i y_k) t} &= 0. \end{aligned}$$

Since $\{t^j e^{\lambda_i t}, t^l e^{(x_k + i y_k)t}\}$ is linearly independent (by using linear algebra fact which is stated), we get that $a_{i,j} = 0$ and $c_{k,l} - i b_{k,l} = 0$ for all $1 \leq i \leq m, 0 \leq j \leq m_i$ and $1 \leq k \leq n, 0 \leq l \leq n_k$. Hence $a_{i,j} = b_{k,l} = c_{k,l} = 0$. Therefore $\{u_{i,j}(t), v_{k,l}(t), w_{k,l}(t)\}$ is linearly independent set.

1.9 Let u_1, u_2 be two linearly independent solutions of the ODE

$$u''(t) + a(t)u'(t) + b(t)u(t) = 0, \quad t \in I$$

where I is an open interval in $\mathbb{R}$ and a, b are continuous functions defined on I . Show that the Wronskian of u_1 and u_2 satisfies the first order ODE

$$W'(t) + a(t)W(t) = 0, \quad t \in I.$$

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1.10 Find the general solutions to the following ODEs.

- (a) $u''(t) - 9u'(t) + 20u(t) = 0, \quad t \in \mathbb{R}.$
- (b) $u''(t) + 4u(t) = \cos t + \sin t, \quad t \in \mathbb{R}.$
- (c) $u'''(t) + 3u''(t) + 3u'(t) + u(t) = 0, \quad t \in \mathbb{R}.$
- (d) $u'''(t) + 3u''(t) + 3u'(t) + u(t) = e^{-t}, \quad t \in \mathbb{R}.$
- (e) $u''''(t) - 16u(t) = 0, \quad t \in \mathbb{R}.$
- (f) $u'''(t) - 16u'(t) = \cos t, \quad t \in \mathbb{R}.$

Ordinary Differential Equations

ASSIGNMENT 2 – MA 417-2025

TOPIC : POWER SERIES METHOD FOR ODES.

21.08.2025

1.11 Find two linearly independent solutions of the form $\sum_{k=0}^{\infty} c_k t^k$ for the following ODEs:

- (a) $u''(t) - tu'(t) + u(t) = 0.$
- (b) $u''(t) + t^3 u'(t) + t^2 u(t) = 0.$

Also, find for what values of t the series converges.

Solution. (a) Let $u(t) = \sum_{k \geq 0} c_k t^k$ satisfy the ODE

$$u''(t) - tu'(t) + u(t) = 0.$$

From the power series, we have

$$u'(t) = \sum_{k=1}^{\infty} k c_k t^{k-1} = \sum_{k=0}^{\infty} (k+1) c_{k+1} t^k,$$

and

$$u''(t) = \sum_{k=2}^{\infty} k(k-1) c_k t^{k-2} = \sum_{k=0}^{\infty} (k+1)(k+2) c_{k+2} t^k.$$

Substituting these into the ODE yields

$$\sum_{k=0}^{\infty} (k+1)(k+2) c_{k+2} t^k - t \sum_{k=1}^{\infty} k c_k t^{k-1} + \sum_{k=0}^{\infty} c_k t^k = 0,$$

which simplifies to

$$\sum_{k=0}^{\infty} [(k+1)(k+2)c_{k+2} - (k-1)c_k] t^k = 0.$$

Thus, $u(t)$ is a solution if and only if

$$c_{k+2} = \frac{k-1}{(k+1)(k+2)} c_k, \quad \forall k \geq 0.$$

This implies

$$c_{2k} = \frac{-1 \cdot 3 \cdot 5 \cdots (2k-3)}{(2k)!} c_0 = \frac{-c_0}{2^k k! (2k-1)}, \quad c_{2k+1} = 0, \quad \forall k \geq 1.$$

Therefore,

$$u(t) = \sum_{k=0}^{\infty} c_{2k} t^{2k} + \sum_{k=0}^{\infty} c_{2k+1} t^{2k+1} = -c_0 \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k! (2k-1)} + c_1 t.$$

Convergence: The series $\sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k! (2k-1)}$ converges absolutely for all $t \in \mathbb{R}$ because

$$\left| \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k! (2k-1)} \right| \leq \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k! |2k-1|} < \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k!} = e^{t^2/2}.$$

Hence,

$$u_1(t) = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k! (2k-1)} \quad \text{and} \quad u_2(t) = t$$

are two linearly independent solutions.

(b) Similarly, let $u(t) = \sum_{k \geq 0} c_k t^k$ satisfy

$$u''(t) + t^3 u'(t) + t^2 u(t) = 0.$$

Substituting the series expansions, we get

$$\sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2} t^k + t^3 \sum_{k=1}^{\infty} k c_k t^{k-1} + t^2 \sum_{k=0}^{\infty} c_k t^k = 0,$$

which can be rewritten as

$$\sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2} t^k + \sum_{k=2}^{\infty} (k-2)c_{k-2} t^k + \sum_{k=2}^{\infty} c_{k-2} t^k = 0.$$

Grouping terms by powers of t ,

$$2c_2 + 6c_3 t + \sum_{k=2}^{\infty} [(k+1)(k+2)c_{k+2} + (k-1)c_{k-2}] t^k = 0.$$

Thus, $u(t)$ solves the ODE if and only if

$$c_2 = c_3 = 0, \quad c_{k+2} = \frac{k-1}{(k+1)(k+2)} c_{k-2}, \quad \forall k \geq 2.$$

This gives

$$c_m = \begin{cases} c_0 \prod_{j=1}^k \frac{4j-3}{(4j-1)4j}, & m = 4k, k \geq 1, \\ c_1 \prod_{j=1}^k \frac{4j-2}{(4j+1)4j}, & m = 4k+1, k \geq 1, \\ 0, & m = 4k+2, k \geq 1, \\ 0, & m = 4k+3, k \geq 1. \end{cases}$$

Therefore,

$$u(t) = c_0 \left(1 + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j-3}{(4j-1)4j} t^{4k} \right) + c_1 \left(t + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j-2}{(4j+1)4j} t^{4k+1} \right).$$

Convergence: Define

$$u_1(t) = 1 + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j-3}{(4j-1)4j} t^{4k}, \quad u_2(t) = t + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j-2}{(4j+1)4j} t^{4k+1}.$$

For u_1 ,

$$\left| \frac{a_{k+1}}{a_k} \right| = \frac{4(k+1)-3}{4(k+1)(4(k+1)-1)} |t|^4 = \frac{4/k + 1/k^2}{4(1+1/k)(4+3/k)} |t|^4 \rightarrow 0 \quad \text{as } k \rightarrow \infty,$$

and similarly for u_2 ,

$$\left| \frac{a_{k+1}}{a_k} \right| = \frac{4(k+1)-2}{4(k+1)(4(k+1)+1)} |t|^4 = \frac{4/k + 2/k^2}{4(1+1/k)(4+5/k)} |t|^4 \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Hence, $u(t)$ converges absolutely for all $t \in \mathbb{R}$.

1.12 Find the solution of the ODE

$$(1+t^2)u''(t) + u(t) = 0, \quad u(0) = 1, \quad u'(0) = 1$$

of the form $\sum_{k=0}^{\infty} c_k t^k$.

Solution. Let $u(t) = \sum_{k=0}^{\infty} c_k t^k$ be a solution to the ODE. From previous computations, we have

$$(1+t^2) \sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2} t^k + \sum_{k=0}^{\infty} c_k t^k = 0,$$

which implies

$$\sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2} t^k + \sum_{k=0}^{\infty} (k-1)kc_k t^k + \sum_{k=0}^{\infty} c_k t^k = 0,$$

and hence

$$\sum_{k=0}^{\infty} [(k+1)(k+2)c_{k+2} + (k^2 - k + 1)c_k] t^k = 0.$$

It follows that $u(t)$ is a solution to the given ODE if and only if

$$c_{k+2} = \frac{-(k^2 - k + 1)}{(k+1)(k+2)} c_k, \quad \forall k \geq 0,$$

which leads to

$$c_{2k} = c_0 (-1)^k \prod_{j=0}^{k-1} \frac{4j^2 - 2j + 1}{(2j+1)(2j+2)}, \quad c_{2k+1} = c_1 (-1)^k \prod_{j=1}^k \frac{4j^2 - 6j + 3}{2j(2j+1)}; \quad \forall k \geq 1.$$

Therefore,

$$\begin{aligned} u(t) &= \sum_{k=0}^{\infty} c_{2k} t^{2k} + \sum_{k=0}^{\infty} c_{2k+1} t^{2k+1} \\ &= c_0 \left[1 + \sum_{k=1}^{\infty} \prod_{j=0}^{k-1} \frac{4j^2 - 2j + 1}{(2j+1)(2j+2)} (-1)^k t^{2k} \right] + c_1 \left[t + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j^2 - 6j + 3}{2j(2j+1)} (-1)^k t^{2k+1} \right]. \end{aligned}$$

Using $u(0) = 1 \implies c_0 = 1$ and $u'(0) = 1 \implies c_1 = 1$, the unique solution to the ODE is

$$u(t) = \left(1 + \sum_{k=1}^{\infty} \prod_{j=0}^{k-1} \frac{4j^2 - 2j + 1}{(2j+1)(2j+2)} (-1)^k t^{2k} \right) + \left(t + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j^2 - 6j + 3}{2j(2j+1)} (-1)^k t^{2k+1} \right).$$

Convergence: Define

$$u_1(t) = 1 + \sum_{k=1}^{\infty} \prod_{j=0}^{k-1} \frac{4j^2 - 2j + 1}{(2j+1)(2j+2)} (-1)^k t^{2k}, \quad u_2(t) = t + \sum_{k=1}^{\infty} \prod_{j=1}^k \frac{4j^2 - 6j + 3}{2j(2j+1)} (-1)^k t^{2k+1}.$$

By the ratio test applied to u_1 , we have

$$\left| \frac{a_{k+1}}{a_k} \right| = \frac{4k^2 - 2k + 1}{(2k+1)(2k+2)} |t|^2 = \frac{4 - \frac{2}{k} + \frac{1}{k^2}}{(2 + \frac{1}{k})(2 + \frac{2}{k})} |t|^2 \rightarrow |t|^2 \quad \text{as } k \rightarrow \infty.$$

Hence, for $|t| < 1$, u_1 converges absolutely. By a similar argument, u_2 converges absolutely for $|t| < 1$, implying that $u(t)$ converges absolutely on $|t| < 1$.

At $|t| = 1$, for $u_1(t)$,

$$\left| \frac{a_{k+1}}{a_k} \right| = \frac{4k^2 - 2k + 1}{(2k+1)(2k+2)} = 1 - \frac{8k+1}{(2k+1)(2k+2)}.$$

Since $k \geq 1$, it follows that $0 < \frac{8k+1}{(2k+1)(2k+2)} < 1$, and thus

$$\log \left| \frac{a_{k+1}}{a_k} \right| = -\frac{8k+1}{(2k+1)(2k+2)} + \mathcal{O}\left(\frac{1}{k^2}\right).$$

Summing from $k = 1$ to $n - 1$ for some $n > 1$ gives

$$\begin{aligned} \log \left| \frac{a_n}{a_1} \right| &= -\sum_{k=1}^{n-1} \frac{8k+1}{(2k+1)(2k+2)} + \mathcal{O}\left(\sum_{k=1}^{n-1} \frac{1}{k^2}\right) \\ &= -\sum_{k=1}^{n-1} \frac{2}{k+1} + \sum_{k=1}^{n-1} \frac{3}{(2k+1)(2k+2)} + \mathcal{O}\left(\sum_{k=1}^{n-1} \frac{1}{k^2}\right) \\ &\leq -2 \sum_{k=1}^n \frac{1}{k} + 3 \sum_{k=0}^{\infty} \frac{1}{(k+1)^2} + \mathcal{O}(1) \\ &\leq -2 \log n + \mathcal{O}(1). \end{aligned}$$

This implies

$$|a_n| \leq \frac{|a_1| e^{\mathcal{O}(1)}}{n^2} \implies |a_n| = \mathcal{O}\left(\frac{1}{n^2}\right) \implies \sum_{k=1}^{\infty} |a_k| = \mathcal{O}\left(\sum_{k=1}^{\infty} \frac{1}{k^2}\right) = \mathcal{O}(1).$$

Thus, $u_1(1)$ converges absolutely. Similarly, $u_2(1)$ also converges absolutely, and therefore $u(1)$ converges absolutely.

Hence, the series solution $u(t)$ converges absolutely for all $t \in [-1, 1]$.

1.13 Consider the ODE

$$(1 - t^2)u''(t) - tu'(t) + \alpha^2 u(t) = 0,$$

where α is a constant.

- (a) Find two linearly independent power series solutions of this equation in $(-1, 1)$.
- (b) Show that for every non-negative integer $\alpha = n$, there is a polynomial solution of degree n .

Remark: The above ODE is called the *Chebyshev equation* and the suitably normalized polynomial solution when $\alpha = n$ is called the *Chebyshev Polynomial*.

Solution. Let $u(t) = \sum_{k=0}^{\infty} c_k t^k$ be a power series solution to the ODE

$$(1 - t^2)u''(t) - tu'(t) + \alpha^2 u(t) = 0,$$

where α is a constant.

Substituting the series into the ODE yields

$$\begin{aligned} 0 &= (1 - t^2) \sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2}t^k - t \sum_{k=1}^{\infty} k c_k t^{k-1} + \alpha^2 \sum_{k=0}^{\infty} c_k t^k \\ &= \sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2}t^k - t^2 \sum_{k=2}^{\infty} (k-1)k c_k t^{k-2} - \sum_{k=0}^{\infty} (k - \alpha^2)c_k t^k \\ &= \sum_{k=0}^{\infty} [(k+1)(k+2)c_{k+2} - (k^2 - \alpha^2)c_k] t^k. \end{aligned}$$

Therefore, the power series $u(t)$ is a solution to the ODE if and only if

$$c_{k+2} = \frac{k^2 - \alpha^2}{(k+1)(k+2)} c_k, \quad \forall k \geq 0.$$

From this recurrence relation, we obtain

$$\begin{aligned} c_2 &= \frac{-\alpha^2 c_0}{2}, \\ c_4 &= \frac{2^2 - \alpha^2}{3 \cdot 4} c_2 = \frac{(2^2 - \alpha^2)(-\alpha^2)c_0}{4!}, \\ c_{2k} &= \frac{(2k-2)^2 - \alpha^2}{(2k-1)2k} c_{2k-2} = \cdots = \frac{c_0}{(2k)!} \prod_{j=0}^{k-1} (4j^2 - \alpha^2), \quad \forall k \geq 1, \end{aligned}$$

and similarly,

$$\begin{aligned} c_3 &= \frac{1 - \alpha^2}{2 \cdot 3} c_1, \\ c_{2k+1} &= \frac{(2k-1)^2 - \alpha^2}{2k(2k+1)} c_{2k-1} = \cdots = \frac{c_1}{(2k+1)!} \prod_{j=1}^k ((2j-1)^2 - \alpha^2), \quad \forall k \geq 1. \end{aligned}$$

Hence, the general solution to the ODE is given by

$$u(t) = c_0 \left(1 + \sum_{k=1}^{\infty} \frac{t^{2k}}{(2k)!} \prod_{j=0}^{k-1} (4j^2 - \alpha^2) \right) + c_1 \left(t + \sum_{k=1}^{\infty} \frac{t^{2k+1}}{(2k+1)!} \prod_{j=1}^k ((2j-1)^2 - \alpha^2) \right).$$

Define

$$u_1(t) = 1 + \sum_{k=1}^{\infty} \frac{t^{2k}}{(2k)!} \prod_{j=0}^{k-1} (4j^2 - \alpha^2), \quad u_2(t) = t + \sum_{k=1}^{\infty} \frac{t^{2k+1}}{(2k+1)!} \prod_{j=1}^k ((2j-1)^2 - \alpha^2),$$

which form two linearly independent solutions to the ODE.

Convergence: Note that

$$\left| \frac{c_{k+2}t^{k+2}}{c_k t^k} \right| = \frac{|k^2 - \alpha^2|}{|(k+1)(k+2)|} \cdot |t|^2 = \frac{\left|1 - \frac{\alpha^2}{k^2}\right|}{\left|(1 + \frac{1}{k})(1 + \frac{2}{k})\right|} \cdot |t|^2 \rightarrow |t|^2 \quad \text{as } k \rightarrow \infty.$$

Thus, by the ratio test, both u_1 and u_2 converge absolutely for $|t| < 1$. Moreover, by the same argument as in the previous problem, $u(t)$ converges absolutely for all $t \in [-1, 1]$.

Suppose now that $\alpha \in \mathbb{N}$. First, if $\alpha = 2n$, then

$$c_{2n+2} = 0, \quad c_{2k} = 0, \quad \forall k \geq n+1,$$

so $u_1(t)$ reduces to a polynomial of degree $2n$.

If $\alpha = 2n+1$, then

$$c_{2n+3} = 0, \quad c_{2k+1} = 0, \quad \forall k \geq n+1,$$

and hence $u_2(t)$ is a polynomial of degree $2n+1$.

Therefore, when α is a nonnegative integer, there always exists a polynomial solution of degree α that satisfies the given ODE.

1.14 Consider the ODE

$$u''(t) - 2tu'(t) + 2\alpha u(t) = 0,$$

where α is a constant.

- (a) Find two linearly independent power series solutions of this equation in $\mathbb{R}$.
- (b) Show that for every non-negative integer $\alpha = n$, there is a polynomial solution of degree n .
- (c) Show that for $\alpha = n$, the function H_n defined by

$$H_n(t) = (-1)^n e^{t^2} \frac{d^n}{dt^n} (e^{-t^2})$$

is a polynomial solution of degree n .

Remark: The above ODE is called the *Hermite equation* and the H_n 's are called the *Hermite Polynomials*.

Solution. Consider the power series solution

$$u(t) = \sum_{k=0}^{\infty} c_k t^k$$

to the above ODE. Similarly as before, substituting the series for u'' , u' , u into the ODE, we observe that

$$\begin{aligned} 0 &= \sum_{k=0}^{\infty} (k+1)(k+2)c_{k+2}t^k - 2t \sum_{k=1}^{\infty} k c_k t^{k-1} + 2\alpha \sum_{k=0}^{\infty} c_k t^k \\ &= \sum_{k=0}^{\infty} [(k+1)(k+2)c_{k+2} - 2(k-\alpha)c_k] t^k. \end{aligned}$$

Therefore, the power series is a solution to the ODE if and only if

$$c_{k+2} = \frac{2(k-\alpha)}{(k+1)(k+2)} c_k, \quad \forall k \geq 0.$$

From this, we obtain that

$$c_2 = \frac{-2\alpha c_0}{1 \cdot 2} = \frac{c_0}{2!} \cdot (-2\alpha),$$

$$c_{2k} = \frac{c_0 2^k}{(2k)!} \prod_{j=0}^{k-1} (j - \alpha), \quad \forall k \geq 1,$$

and similarly,

$$c_3 = \frac{2(1-\alpha)c_1}{2 \cdot 3} = \frac{c_1}{3!} \cdot 2(1-\alpha),$$

$$c_{2k+1} = \frac{c_1 2^k}{(2k+1)!} \prod_{j=1}^k (2j-1-\alpha), \quad \forall k \geq 1.$$

Hence,

$$u(t) = c_0 \left(1 + \sum_{k=1}^{\infty} \frac{2^k t^{2k}}{(2k)!} \prod_{j=0}^{k-1} (j - \alpha) \right) + c_1 \left(t + \sum_{k=1}^{\infty} \frac{2^k t^{2k+1}}{(2k+1)!} \prod_{j=1}^k (2j-1-\alpha) \right),$$

so

$$u_1(t) = 1 + \sum_{k=1}^{\infty} \frac{2^k t^{2k}}{(2k)!} \prod_{j=0}^{k-1} (j - \alpha), \quad u_2(t) = t + \sum_{k=1}^{\infty} \frac{2^k t^{2k+1}}{(2k+1)!} \prod_{j=1}^k (2j-1-\alpha)$$

are two linearly independent solutions.

Convergence: Note that

$$\left| \frac{c_{k+2} t^{k+2}}{c_k t^k} \right| = \frac{2|k-\alpha|}{(k+1)(k+2)} |t|^2 = \frac{|2/k - \alpha/k^2| |t|^2}{|(1+1/k)(1+2/k)|} \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Therefore, both u_1, u_2 have radius of convergence $R = \infty$ and converge absolutely for all $t \in (-\infty, +\infty)$.

If $\alpha \in \mathbb{N}$, then if $\alpha = 2n$, we have $c_{2(n+1)} = 0$ and $c_{2k} = 0$ for all $k \geq n+1$. Thus, $u_1(t)$ is a polynomial of degree $2n$. If $\alpha = 2n-1$, then $c_{2n+1} = 0$ and $c_{2k+1} = 0$ for all $k \geq n$, so $u_2(t)$ is a polynomial of degree $2n-1$.

Next, define

$$H_n(t) = (-1)^n e^{t^2} \frac{d^n}{dt^n} e^{-t^2}.$$

We shall show that H_n satisfies the given ODE and that H_n is a polynomial. To reduce notation, write $\frac{d}{dt} = D$.

Then,

$$H_n = (-1)^n e^{t^2} D^n e^{-t^2},$$

which implies

$$H'_n = (-1)^n \left(2te^{t^2} D^n e^{-t^2} + e^{t^2} D^{n+1} e^{-t^2} \right) = 2tH_n - H_{n+1}.$$

Differentiating again,

$$H''_n = D(2tH_n - H_{n+1}) = 2H_n + 2tH'_n - H'_{n+1},$$

which yields

$$H''_n - 2tH'_n + 2nH_n = 2(n+1)H_n - H'_{n+1}. \quad (1)$$

To compute H'_{n+1} , note

$$\begin{aligned} H'_{n+1} &= D \left((-1)^{n+1} e^{t^2} D^{n+1} e^{-t^2} \right) \\ &= (-1)^{n+1} \left[2te^{t^2} D^{n+1} e^{-t^2} + e^{t^2} D^{n+2} e^{-t^2} \right] \\ &= 2tH_{n+1} + (-1)^{n+1} e^{t^2} \sum_{k=0}^{n+1} \binom{n+1}{k} D^k (-2t) D^{n+1-k} e^{-t^2}. \end{aligned}$$

Since $D^k(-2t) = 0$ for all $k \geq 2$, this reduces to

$$\begin{aligned} H'_{n+1} &= 2tH_{n+1} + (-1)^{n+1}e^{t^2} \left(-2tD^{n+1}e^{-t^2} - 2(n+1)D^n e^{-t^2} \right) \\ &= 2tH_{n+1} - 2tH_{n+1} + 2(n+1)(-1)^n e^{t^2} D^n e^{-t^2} = 2(n+1)H_n. \end{aligned} \quad (2)$$

Substituting (2) into (1) shows

$$H''_n - 2tH'_n + 2nH_n = 0,$$

so H_n satisfies the ODE.

Moreover,

$$2tH_n - H_{n+1} = H'_n = 2nH_{n-1}$$

by (2). Hence the recurrence

$$H_{n+1} = 2tH_n - 2nH_{n-1}, \quad \forall n \geq 1.$$

Also,

$$H_0(t) = (-1)^0 e^{t^2} D^0 e^{-t^2} = 1,$$

and

$$H_1(t) = (-1)^1 e^{t^2} D e^{-t^2} = -e^{t^2} (-2te^{-t^2}) = 2t.$$

Therefore,

$$H_2 = 2tH_1 - 2H_0 = 4t^2 - 2,$$

a polynomial. By induction, H_n is a polynomial of degree n for all n .

1.15 Find all solutions of the ODE

$$t^2 u''(t) + 2tu'(t) - 6u(t) = 0, \quad t > 0.$$

Solution. Let $u(t) = t^\lambda$ be a solution to the above differential equation. Computing derivatives, we find

$$Du(t) = \lambda t^{\lambda-1}, \quad D^2u(t) = \lambda(\lambda-1)t^{\lambda-2}.$$

Substituting into the equation yields

$$t^\lambda [\lambda(\lambda-1) + 2\lambda - 6] = 0,$$

which simplifies (since $t^\lambda \neq 0$ for $t > 0$) to

$$\lambda^2 + \lambda - 6 = 0 \quad \Rightarrow \quad \lambda = 2, -3.$$

Thus, the general solution is

$$u(t) = at^2 + \frac{b}{t^3}, \quad \forall t > 0,$$

where $a, b \in \mathbb{R}$.

1.16 Find all solutions of the form

$$u(t) = |t|^r \sum_{k=0}^{\infty} c_k t^k$$

for the ODE

$$t^2 u''(t) + tu'(t) + \left(t^2 - \frac{1}{4} \right) u(t) = 0, \quad |t| > 0.$$

Solution. Wlog, assume $u(t) = t^r v(t)$; where r is a constant to be determined later, be a solution to the following ODE

$$t^2 u''(t) + tu'(t) + (t^2 - \frac{1}{4})u(t) = 0$$

where $t > 0$. Then we see that

$$u'(t) = rt^{r-1}v(t) + t^r v'(t), \quad u''(t) = t^{r-2} (t^2 v''(t) + 2rtv'(t) + (r^2 - r)v(t))$$

Now employing these results into the ODE, we see that $t^r v(t)$ is a solution iff

$$\begin{aligned} 0 &= t^r (t^2 v''(t) + 2rtv'(t) + (r^2 - r)v(t)) + t^r (rv(t) + tv'(t)) + t^r \left(t^2 - \frac{1}{4}\right) v(t) \\ &= t^r \left[t^2 v''(t) + (2r + 1)tv'(t) + \left(r^2 + t^2 - \frac{1}{4}\right) v(t) \right] \end{aligned}$$

Since $t > 0$, we see

$$t^2 v''(t) + (2r + 1)tv'(t) + \left(r^2 + t^2 - \frac{1}{4}\right) v(t) = 0 \quad (4)$$

Now let $v(t) = \sum_{k=0}^{\infty} c_k t^k$ be a solution to the ODE in (4). Then we see from above computations, it is so iff

$$\begin{aligned} 0 &= t^2 \sum_{k=2}^{\infty} k(k-1)c_k t^{k-2} + (2r+1)t \sum_{k=1}^{\infty} k c_k t^{k-1} + \left(r^2 + t^2 - \frac{1}{4}\right) \sum_{k=0}^{\infty} c_k t^k \\ &= \sum_{k=0}^{\infty} \left[k(k-1)c_k + (2r+1)k c_k + \left(r^2 - \frac{1}{4}\right)c_k \right] t^k + \sum_{k=0}^{\infty} c_k t^{k+2} \\ &= \sum_{k=0}^{\infty} \left[(k^2 + 2rk + r^2 - \frac{1}{4})c_k \right] t^k + \sum_{k=2}^{\infty} c_{k-2} t^k \\ &= (r^2 - \frac{1}{4})c_0 + (r^2 + 2r + \frac{3}{4})c_1 t + \sum_{k=2}^{\infty} \left[(k^2 + 2rk + r^2 - \frac{1}{4})c_k + c_{k-2} \right] t^k \end{aligned}$$

Therefore we see the proposed series is a solution to the ODE in (4) iff

$$\left(r^2 - \frac{1}{4}\right) c_0 = 0 \quad (5)$$

$$\left(r^2 + 2r + \frac{3}{4}\right) c_1 = 0 \quad (6)$$

$$(k^2 + 2rk + r^2 - \frac{1}{4}) c_k + c_{k-2} = 0; \quad \forall k \geq 2 \quad (7)$$

Suppose first $c_0 \neq 0, c_1 \neq 0$ then the equations above force that $r = \frac{-1}{2}$. Then we see $k^2 + 2rk + r^2 - \frac{1}{4} = k^2 - k = k(k-1) \neq 0$ since $k \geq 2$. Therefore we see then,

$$c_k = \frac{-c_{k-2}}{k(k-1)}; \quad \forall k \geq 2$$

Which implies that

$$c_2 = \frac{-c_0}{2!}, c_4 = \frac{-c_2}{3 \cdot 4} = \frac{c_0}{4!}, \dots, c_{2k} = \frac{(-1)^k c_0}{(2k)!}; \quad \forall k \geq 1$$

and similarly

$$c_3 = \frac{-c_1}{3!}, c_5 = \frac{-c_3}{4 \cdot 5} = \frac{c_1}{5!}, \dots, c_{2k+1} = \frac{(-1)^k c_1}{(2k+1)!}; \quad \forall k \geq 1$$

Therefore we see that

$$v(t) = c_0 \left(1 + \sum_{k=1}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!} \right) + c_1 \left(t + \sum_{k=1}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!} \right)$$

Now we notice that $v_1(t) = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!} = \cos t$ and $v_2(t) = t + \sum_{k=1}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!} = \sin t$, therefore we see

$$v(t) = c_0 \cos t + c_1 \sin t$$

Since $\sin t, \cos t$ are analytic everywhere, so $v(t)$ is analytic everywhere, and thus $v(t)$ converges everywhere. Therefore in this case we see

$$u(t) = \frac{v(t)}{\sqrt{t}} = \frac{c_0 \cos t + c_1 \sin t}{\sqrt{t}}$$

converges for $t > 0$. Now suppose $c_0 \neq 0, c_1 = 0$, then we see that $c_0 \neq 0 \implies r = \pm \frac{1}{2} \implies k^2 + 2rk + r^2 - \frac{1}{4} = k(k \pm 1) \neq 0$ since $k \geq 2$. Hence in this case we see $c_k = \frac{-c_{k-2}}{k(k \pm 1)}$ according to $r = \pm \frac{1}{2}$. So when $r = \frac{1}{2}$, we see $c_{2k} = \frac{(-1)^k c_0}{(2k+1)!}$ and $c_{2k+1} = 0$ for all $k \geq 0$. Hence in this case we see $v(t) = c_0 \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k+1)!}$. Then we see

$$u(t) = c_0 \sqrt{t} \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k+1)!} = \frac{c_0 \sin t}{\sqrt{t}}$$

and this converges for $t > 0$ since $v(t)$ has infinite radius of convergence. Similarly if $r = -\frac{1}{2}$ we see $c_{2k} = \frac{(-1)^k c_0}{(2k)!}, c_{2k+1} = 0$ for all $k \geq 0$ and thus

$$u(t) = \frac{c_0 \cos t}{\sqrt{t}}$$

Now we consider the case when $c_0 = 0, c_1 \neq 0$ this forces us to see $r = -\frac{1}{2}, -\frac{3}{2}$. Now when $r = -\frac{3}{2} \implies k^2 + 2rk + r^2 - \frac{1}{4} = k^2 - 3k + 2 = (k-2)(k-1) \neq 0$ when $k > 2$ is an integer. Therefore we have $c_k = \frac{-c_{k-2}}{(k-1)(k-2)}; \forall k \geq 3$ which implies

$$c_0 = 0, c_1 \neq 0, c_3 = \frac{-c_1}{2!}, c_4 = \frac{-c_2}{3!}, c_5 = \frac{(-1)^2 c_1}{4!}, c_6 = \frac{(-1)^2 c_2}{5!}, \dots$$

where c_2 is may or may not be zero. Therefore we see in this case

$$\begin{aligned} v(t) &= 0 + c_1 t + c_2 t^2 + \frac{-c_1 t^3}{2!} + \frac{-c_2 t^4}{3!} + \frac{(-1)^2 c_1 t^5}{4!} + \frac{(-1)^2 c_2 t^6}{5!} + \dots \\ &= c_1 t \cos t + c_2 t \sin t \end{aligned}$$

Hence

$$u(t) = t^{-\frac{3}{2}} v(t) = \frac{c_1 \cos t + c_2 \sin t}{\sqrt{t}}$$

Now if $r = -\frac{1}{2}$ we get the same solution as before

$$u(t) = \frac{c_1 \sin t}{\sqrt{t}}$$

Finally we consider the case where $c_0 = c_1 = 0$. Then we see r can take any value. Also we see $k^2 + 2rk + r^2 - \frac{1}{4} = 0 \iff k = \pm \frac{1}{2} - r$. Therefore if $\pm \frac{1}{2} - r$ is not an integer ≥ 2 , we see simply $c_k = \frac{-c_{k-2}}{(k+r)^2 - \frac{1}{4}} \implies c_k = 0; \forall k \geq 0$ so there are no non-trivial solutions. Thus we assume $\pm \frac{1}{2} - r$ are integers ≥ 2 . Note that $-\frac{1}{2} - r = m, \frac{1}{2} - r = m+1$ then we see

$$c_k = \frac{-c_{k-2}}{(k-m)(k-m-1)}; \forall k \geq 2, k \neq m, k \neq m+1$$

Then, $c_0 = c_1 = 0, c_2 = 0, \dots, c_{m-1} = 0, c_{m+2} = \frac{-c_m}{2!}, c_{m+3} = \frac{-c_{m+1}}{3!}, c_{m+4} = \frac{(-1)^2 c_m}{4!} \dots$, where c_m, c_{m+1} are any constants. Thus in this case

$$v(t) = c_m t^m \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{2k!} + c_{m+1} t^m \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!} = t^m c_m \cos t + t^m c_{m+1} \sin t$$

Hence

$$u(t) = t^r v(t) = t^{-\frac{1}{2}-m} \cdot t^m (c_m \cos t + c_{m+1} \sin t) = \frac{c_m \cos t + c_{m+1} \sin t}{\sqrt{t}}$$

Therefore we see from the above analysis that all solutions of the given ODE are of the form

$$\boxed{\frac{A \cos t + B \sin t}{\sqrt{|t|}}; |t| > 0}$$

for any constants A, B .

1.17 Consider the ODE

$$tu''(t) + (1-t)u'(t) + \alpha u(t) = 0,$$

where α is a constant.

- (a) Show that this equation has a regular singular point at $t = 0$.
- (b) Compute the indicial polynomial and its roots.
- (c) Find a solution u of the form

$$u(t) = |t|^r \sum_{k=0}^{\infty} c_k t^k, \quad |t| > 0.$$

- (d) Show that for every non-negative integer $\alpha = n$, there is a polynomial solution of degree n .
- (e) Show that for $\alpha = n$, the function L_n defined by

$$L_n(t) = e^t \frac{d^n}{dt^n} (t^n e^{-t})$$

is a polynomial solution of degree n .

Remark: The above ODE is called the *Laguerre equation* and the L_n 's are called the *Laguerre Polynomials*.

Solution. (a) If we write the given ODE in the form $u'' + p(t)u' + q(t)u = 0$ then we see that $p(t) = \frac{1-t}{t}$, $q(t) = \frac{\alpha}{t}$. Now we see that $\lim_{t \rightarrow 0} p(t) = \infty$ and $\lim_{t \rightarrow 0} q(t) = \infty$ therefore, $t = 0$ is a singular point of this ODE. Now

$$\lim_{t \rightarrow 0} tp(t) = \lim_{t \rightarrow 0} (1-t) = 1 \text{ and } \lim_{t \rightarrow 0} t^2 q(t) = \lim_{t \rightarrow 0} \alpha t = 0$$

Since these are finite we see $t = 0$ is actually a regular singular point (non-essential singularity) of the ODE.

- (b) Wlog, let $u(t) = t^r \sum_{k=0}^{\infty} c_k t^k$ be a solution to the ODE given above for $t > 0$. Compute

$$u'(t) = \sum_{k=0}^{\infty} (k+r) c_k t^{k+r-1}, \quad u''(t) = \sum_{k=0}^{\infty} (k+r-1)(k+r) c_k t^{k+r-2}$$

and substitute these series into the ODE,

$$\begin{aligned} 0 &= t \sum_{k=0}^{\infty} (k+r-1)(k+r) c_k t^{k+r-2} + (1-t) \sum_{k=0}^{\infty} (k+r) c_k t^{k+r-1} + \alpha \sum_{k=0}^{\infty} c_k t^{k+r} \\ &= \sum_{k=0}^{\infty} [(k+r-1)(k+r) c_k + (k+r) c_k] t^{k+r-1} + \sum_{k=0}^{\infty} [\alpha c_k - (k+r) c_k] t^{k+r} \\ &= [(r-1)rc_0 + rc_0] t^{r-1} + \sum_{k=0}^{\infty} (k+r+1)^2 c_{k+1} t^{k+r} + \sum_{k=0}^{\infty} (\alpha - k - r) c_k t^{k+r} \\ &= r^2 c_0 t^{r-1} + \sum_{k=0}^{\infty} [(k+r+1)^2 c_{k+1} - (k+r-\alpha) c_k] t^{k+r} \end{aligned} \tag{8}$$

Therefore the indicial polynomial is $I(r) = r^2$ and it has root $r = 0$ with multiplicity 2.

(c) From (8), we see that $t^r \sum_{k \geq 0} c_k t^k$ is a solution to the ODE, iff

$$r^2 c_0 = 0; (k + r + 1)^2 c_{k+1} = (k + r - \alpha) c_k; \forall k \geq 0$$

First we consider $c_0 \neq 0$. Which forces $r = 0$ implying

$$c_{k+1} = \frac{(k - \alpha) c_k}{(k + 1)^2}; \forall k \geq 0$$

Therefore we get

$$c_1 = \frac{-\alpha c_0}{1^2}, c_2 = \frac{(-1)^2 \alpha(\alpha - 1) c_0}{(2!)^2}, \dots, c_k = \frac{(-1)^k \alpha(\alpha - 1) \cdots (\alpha - k + 1) c_0}{(k!)^2}; \forall k \geq 1$$

Thus we have

$$u(t) = c_0 \left[1 + \sum_{k=1}^{\infty} \frac{(-1)^k \alpha(\alpha - 1) \cdots (\alpha - k + 1)}{(k!)^2} t^k \right]$$

Since

$$\left| \frac{c_{k+1} t^{k+1}}{c_k t^k} \right| = \frac{|k - \alpha| |t|}{(k + 1)^2} \rightarrow 0 \text{ as } k \rightarrow \infty$$

we see by ratio test, this solution converges for all $t \in (-\infty, \infty)$.

Now we consider the case when $c_0 = 0$. Then r can take any value. Suppose $r \geq 0$ then we see $(k + r + 1)^2 > 0$ and hence $c_{k+1} = \frac{k+r-\alpha}{(k+r+1)^2} c_k \implies c_k = 0$ for all k . If $r < 0$, then we consider two cases. First r is not an integer. Then we see $(k + r + 1)^2 > 0$ and consequently $c_k = 0$ for all k . Next we consider if $r = -n$ for some $n \in \mathbb{N}$, then $(k + r + 1)^2 = 0$ for only $k = n - 1$. Therefore we see

$$c_{k+1} = \frac{k - n - \alpha}{(k - n + 1)^2} c_k; \forall k \geq 0, k \neq n - 1$$

Which implies $c_0 = c_1 = \dots = c_{n-1} = 0$ and $c_{k+1} = \frac{k-n-\alpha}{(k-n+1)^2} c_k; \forall k \geq n$, where c_n is a constant may or may not be zero. Therefore we see

$$c_{n+1} = \frac{-\alpha c_n}{1^2}, c_{n+2} = \frac{(-1)^2 \alpha(\alpha - 1) c_n}{(2!)^2}, \dots, c_{n+k} = \frac{(-1)^k \alpha(\alpha - 1) \cdots (\alpha - k + 1) c_n}{(k!)^2}; \forall k \geq 1$$

Which again gives the same solution as before just c_0 is replaced by c_n .

★ Since we know that the solution space of a second order linear ODE is 2-dimensional, there must be another solution which is linearly independent to the one we have found till now. Since this wasn't asked to be produced, we leave it here.

(d) Let $\alpha = n$ where n is a non-negative integer. Then by part-(c), we see that

$$\begin{aligned} u(t) &= c_0 \left[1 + \sum_{k=1}^{\infty} \frac{(-1)^k \alpha(\alpha - 1) \cdots (\alpha - k + 1)}{(k!)^2} t^k \right] \\ &= c_0 \left[1 + \sum_{k=1}^{\infty} \frac{(-1)^k n(n - 1) \cdots (n - k + 1)}{(k!)^2} t^k \right] \\ &= c_0 \left[1 + \sum_{k=1}^{\infty} \frac{(-1)^k n!}{(n - k)! (k!)^2} t^k \right] \\ &= c_0 \sum_{k=0}^{\infty} \binom{n}{k} \frac{(-1)^k t^k}{k!} \\ &= c_0 \sum_{k=0}^n \binom{n}{k} \frac{(-1)^k t^k}{k!} \end{aligned} \tag{9}$$

Clearly (9) is a polynomial of degree n , since $\binom{n}{k} = 0$ for all $k \geq n + 1$. Therefore we found our polynomial solution of the ODE.

(e) Define

$$L_n(t) = e^t D^n(t^n e^{-t})$$

We shall show that $L_n(t)$ satisfies the ODE above and is a polynomial of degree n for any $n \in \mathbb{N}$. We see by Leibniz Rule,

$$L_n(t) = e^t \sum_{k=0}^n \binom{n}{k} D^k(t^n) D^{n-k}(e^{-t})$$

and now we notice $D^k t^n = n D^{k-1} t^{n-1} = n(n-1) D^{k-2} t^{n-2} = \dots = n(n-1) \dots (n-k+1) D^0 t^{n-k} = \frac{n! t^{n-k}}{(n-k)!}$ and $D^{n-k} e^{-t} = (-1)^{n-k} e^{-t}$, therefore by plugging these results in L_n , we see

$$\begin{aligned} L_n(t) &= e^t \sum_{k=0}^n \frac{n!}{k!(n-k)!} \cdot \frac{n!}{(n-k)!} \cdot (-t)^{n-k} e^{-t} \\ &= \sum_{k=0}^n \frac{(n!)^2 (-t)^k}{(n-k)!(k!)^2} \\ &= n! \sum_{k=0}^n \binom{n}{k} \frac{(-1)^k t^k}{k!} \\ &= \frac{n! u(t)}{c_0} \end{aligned} \tag{10}$$

(by (9))

Now (10) implies L_n is a polynomial of degree n and since $u(t)$ derived in (9) is a solution to the ODE, we see that $L_n(t) = \frac{n! u(t)}{c_0}$ is a solution too.

1.18 Consider the differential equation:

$$2t(1+t)u''(t) + (3+t)u'(t) - tu(t) = 0.$$

- (a) Check if $t = 0$ and $t = -1$ are regular singular points.
- (b) Find the indicial equations and their roots at both singular points $t = 0$ and $t = -1$.
- (c) Find two linearly independent solutions of the ODE around $t = 0$, i.e., for $0 < |t| < R$, for some $R > 0$.
- (d) Find a solution of the ODE around $t = -1$, i.e., for $0 < |t + 1| < \rho$, for some $\rho > 0$. Can another linearly independent solution be found around $t = -1$?

Solution.

- (a) Rewriting the given ODE as

$$u''(t) + \frac{3+t}{2t(1+t)} u'(t) - \frac{1}{2(1+t)} u(t) = 0,$$

we identify

$$p(t) = \frac{3+t}{2t(1+t)}, \quad q(t) = -\frac{1}{2(1+t)}.$$

Then

$$\lim_{t \rightarrow 0} t p(t) = \frac{3}{2}, \quad \lim_{t \rightarrow 0} t^2 q(t) = 0,$$

so $t = 0$ is a regular singular point. Similarly,

$$\lim_{t \rightarrow -1} (t+1) p(t) = -1, \quad \lim_{t \rightarrow -1} (t+1)^2 q(t) = 0,$$

hence $t = -1$ is also a regular singular point.

(b) At $t = 0$, the indicial equation is

$$r(r-1) + \frac{3}{2}r = 0 \quad \Rightarrow \quad r = 0, -\frac{1}{2}.$$

At $t = -1$, the indicial equation is

$$r(r-1) - r = 0 \quad \Rightarrow \quad r = 0, 2.$$

(c) Let $u(t) = t^r \sum_{k=0}^{\infty} c_k t^k$ be a solution near $t = 0$. Since $tp(t)$ and $t^2q(t)$ are analytic in $(-1, 1)$, the Frobenius expansion converges at least for $|t| < 1$. After substitution we obtain the recurrence

$$(2r+1)rc_0 = 0, \quad (2r-1)rc_0 + (2r+3)(r+1)c_1 = 0,$$

and for $k \geq 0$,

$$(2k+2r+5)(k+r+2)c_{k+2} + (2k+2r+1)(k+r+1)c_{k+1} - c_k = 0.$$

For $r = 0$, we find $c_1 = 0$ and from the recurrence

$$c_2 = \frac{1}{20}c_0, \quad c_3 = -\frac{1}{210}c_0, \quad \dots$$

giving a solution

$$u(t) = c_0 \left(1 + \frac{1}{20}t^2 - \frac{1}{210}t^3 + \dots \right), \quad |t| < 1.$$

For $r = -\frac{1}{2}$, we obtain $c_1 = -c_0$, $c_2 = \frac{1}{6}c_0$, $c_3 = -\frac{1}{10}c_0$, etc., yielding

$$u(t) = |t|^{-\frac{1}{2}} \left(c_0 - c_0 t + \frac{1}{6}c_0 t^2 - \frac{1}{10}c_0 t^3 + \dots \right), \quad 0 < |t| < 1.$$

(d) For $t = -1$, set $t = s - 1$, $\tilde{u}(s) = u(s - 1)$. The equation becomes

$$2(s-1)s\tilde{u}''(s) + (s+2)\tilde{u}'(s) - (s-1)\tilde{u}(s) = 0.$$

Substituting $\tilde{u}(s) = s^r \sum_{k=0}^{\infty} c_k s^k$, we find the indicial equation

$$I(r) = 2r - 2r(r-1) = 0 \quad \Rightarrow \quad r = 0, 2,$$

and recurrence

$$((2k+2r-1)(k+r+1)+1)c_{k+1} - 2(k+r+2)(k+r)c_{k+2} - c_k = 0.$$

For $r = 2$, we get $c_1 = -\frac{7}{6}c_0$ and higher coefficients recursively. Thus

$$\tilde{u}(s) = c_0 s^2 - \frac{7}{6}c_0 s^3 - \frac{69}{120}c_0 s^4 + \dots,$$

giving

$$u_1(t) = (t+1)^2 - \frac{7}{6}(t+1)^3 - \frac{69}{120}(t+1)^4 + \dots$$

For $r = 0$, since $r_1 - r_2 = 2 \in \mathbb{N}$, the second independent solution has the Frobenius-logarithmic form

$$\tilde{u}_2(s) = c \tilde{u}_1(s) \log s + \left(1 + \sum_{k=1}^{\infty} d_k s^k \right),$$

where $d_k = \frac{d}{dr}(rc_k(r))|_{r=0}$ and $c_k(r)$ are from the recurrence. The constant c is uniquely determined by requiring the logarithmic solution to satisfy the ODE (the log terms introduce singularities which cancel only for one choice). A short calculation gives

$$c = -\frac{9}{16}.$$

Thus, reverting to t ,

$$u_2(t) = -\frac{9}{16}u_1(t) \log |t+1| + \left(1 - \frac{1}{2}(t+1) - \frac{1}{8}(t+1)^2 + \dots \right).$$

Since $(t+1)p(t)$ and $(t+1)^2q(t)$ are analytic in $(-2, 0)$, the Frobenius expansions are valid for $0 < |t+1| < 1$.

Ordinary Differential Equations

ASSIGNMENT 3 – MA 417-2025

TOPIC : EXISTENCE AND UNIQUENESS FOR FIRST ORDER ODES.
 10.09.2025

1.1 Discuss the uniqueness and existence of solutions for the following ODEs. Also determine the maximal intervals of existence.

(a) $y' = y^{\frac{1}{2}}, y(0) = 1.$

(b) $y' = y^{\frac{1}{3}}, y(0) = 1.$

Solution. (a) $y(x) = \frac{(x+2)^2}{4}$, unique solution on the maximal interval $(-\infty, \infty).$

(b) $y(x) = \left(\frac{2}{3}x + 1\right)^{3/2}$, unique solution on the maximal interval $[-\frac{3}{2}, \infty).$

1.2 (a) Let σ be a $C^1(\mathbb{R})$ function satisfying the inequality

$$|\sigma'(x)| \leq K\sigma(x), \quad -\infty < x < \infty,$$

where $K > 0$ is a constant. Then show that

$$\sigma(a)e^{-K|x-a|} \leq \sigma(x) \leq \sigma(a)e^{K|x-a|}, \quad \text{for any } a \in (-\infty, +\infty).$$

(b) Consider the IVP

$$y'(x) = f(x, y(x)), \quad y(x_0) = y_0,$$

where $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ satisfies $|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|$ for all $x, y_1, y_2 \in \mathbb{R}$ and is continuous in x . Prove that the IVP has a unique solution. (Hint: Can one choose a suitable σ in (a)?).

Remark. More generally the above IVP will have a unique solution if the function f is *Carathéodory* i.e. f is measurable in x , continuous in y and f is locally bounded satisfying a local *Lipschitz* condition in y .

(c) Suppose the function f in (b) satisfies $|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2||\log|y_1 - y_2||$ and is continuous in x . What can you say about existence and uniqueness of solutions?

Solution. (a) Let $f(x) = \sigma(x)e^{-K(x-a)}$ for $x \geq a$. Then $f'(x) = (\sigma' - K\sigma)e^{-K(x-a)} \leq 0$, so $f(x) \leq f(a) = \sigma(a)$, giving $\sigma(x) \leq \sigma(a)e^{K(x-a)}$. Similarly, $g(x) = \sigma(x)e^{K(x-a)}$ has $g'(x) = (\sigma' + K\sigma)e^{K(x-a)} \geq 0$, so $g(x) \geq g(a) = \sigma(a)$, giving $\sigma(x) \geq \sigma(a)e^{-K(x-a)}$. Repeating for $x \leq a$ yields $\sigma(a)e^{-K|x-a|} \leq \sigma(x) \leq \sigma(a)e^{K|x-a|}$ for any $a \in \mathbb{R}$.

(b) Let $y_1(x)$ and $y_2(x)$ be two solutions and define $\sigma(x) = (y_1(x) - y_2(x))^2$. Then $\sigma \in C^1$ and $\sigma'(x) = 2(y_1 - y_2)(y_1' - y_2') = 2(y_1 - y_2)(f(x, y_1) - f(x, y_2)) \leq 2K(y_1 - y_2)^2 = 2K\sigma(x)$. By the exponential bound from (a), $\sigma(x) \leq \sigma(x_0)e^{2K|x-x_0|} = 0$, since $\sigma(x_0) = (y_1(x_0) - y_2(x_0))^2 = 0$. Hence $\sigma(x) = 0$ for all x , i.e., $y_1(x) = y_2(x)$ for all x , proving uniqueness. Global existence follows by Picard-Lindelöf.

(c) Start with $\sigma(x) = (y_1(x) - y_2(x))^2 \geq 0$. Then $\sigma'(x) = 2(y_1 - y_2)(y_1' - y_2') = 2(y_1 - y_2)(f(x, y_1) - f(x, y_2)) \leq 2K\sigma|\log \sqrt{\sigma}| = K\sigma|\log \sigma|$.

To verify the Osgood condition, let $\phi(u) = u|\log u|$ for $0 < u < 1$. Then $\int_0^1 \frac{du}{\phi(u)} = \int_0^1 \frac{du}{u|\log u|} = \infty$, which diverges. Hence the Osgood criterion applies. Since $\sigma(x_0) = 0$, it follows that $\sigma(x) \equiv 0$, i.e., $y_1(x) = y_2(x)$ for all x , proving uniqueness. By Peano, you only get local existence. In general you don't get global existence unless f satisfies some growth conditions.

- 1.3** Consider the ODE $y' = 3y^{\frac{2}{3}}$ with $y(0) = 0$. Show that the Lipschitz condition is not necessary to have the existence of a solution. Is the solution unique? What happens if $y(0) = 1$?

Solution. First note that $y(x) \equiv 0$ is a solution, so existence is guaranteed even though the Lipschitz condition fails at $y = 0$. Another nontrivial solution is $y(x) = x^3$. In fact, there is an infinite family of nontrivial solutions of the form

$$y(x) = \begin{cases} 0, & x \leq a, \\ (x-a)^3, & x \geq a, \end{cases} \quad a \geq 0,$$

so the solution is not unique when $y(0) = 0$. On the other hand, if $y(0) = 1$, the unique solution is $y(x) = (1+x)^3$, valid on the maximal interval $(-\infty, \infty)$.

- 1.4** Consider the ODE $y' = 1 + y^{\frac{2}{3}}$ with $y(0) = 0$. Show that the Lipschitz condition is not necessary for uniqueness of solution.

Solution. Set $z = y^{1/3}$ so $y = z^3$, $dy = 3z^2 dz$. Then $dx = \frac{dy}{1+y^{2/3}} = \frac{3z^2}{1+z^2} dz$. Integration gives $x = 3(z - \arctan z)$ with $z = y^{1/3}$. Since $\psi(z) = 3(z - \arctan z)$ is continuous, strictly increasing, $\psi(0) = 0$, and $\psi(z) \rightarrow \pm\infty$ as $z \rightarrow \pm\infty$, it is invertible. Thus the unique solution is $y(x) = (\psi^{-1}(x))^3$, valid for all $x \in \mathbb{R}$. Although $f(y) = 1 + y^{2/3}$ is not Lipschitz at $y = 0$, uniqueness holds, showing Lipschitz is sufficient but not necessary.

- 1.5** Prove that the equation

$$y'(t) = \frac{e^{-y^2(t)} \sin^2 t}{1+t^2}, \quad y(0) = 1,$$

has a unique solution. Also show that the maximal interval of existence is $\mathbb{R}$.

Solution. The right-hand side $f(t, y) = \frac{e^{-y^2} \sin^2 t}{1+t^2}$ is continuous in t and y for all $(t, y) \in \mathbb{R}^2$ and is globally Lipschitz in y because $|\partial f / \partial y| = |(-2y)e^{-y^2} \sin^2 t / (1+t^2)| \leq 2|y|e^{-y^2} \leq 1$ (bounded). By the *Picard–Lindelöf* theorem, a unique solution exists locally for the initial value and this local solution can now be uniquely extended to all $t \in \mathbb{R}$. Hence the maximal interval of existence is $\mathbb{R}$.

- 1.6** Can you say anything about the uniqueness and existence of solutions to the following ODE?

$$x' = \begin{cases} x \sin(\frac{1}{x}), & x \neq 0, \\ 0, & x = 0, \end{cases} \quad x(0) = 0.$$

Solution. Consider the function $f(t, x) = \begin{cases} x \sin(\frac{1}{x}), & x \neq 0, \\ 0, & x = 0. \end{cases}$

Note that f has linear growth:

$$|f(t, x)| \leq |x| \quad \forall t, x \in \mathbb{R}.$$

If $x(t)$ solves $x' = f(x)$, then

$$|x(t)| \leq |x(0)| + \int_0^t |x(s)| ds.$$

By Grönwall's inequality,

$$|x(t)| \leq |x(0)|e^{|t|}.$$

Hence no finite-time blow-up occurs, so any solution is global.

Since f is continuous everywhere, Peano's theorem guarantees the existence of a local solution for the IVP. By the above argument, this local solution extends to a global one.

Finally, for the IVP, $x(0) = 0$, and the exponential bound give

$$|x(t)| \leq |x(0)|e^{|t|} = 0,$$

so $x(t) \equiv 0$.

Remark: Note that $f(t, x)$ is not *Lipchitz* in x for any neighbourhood of origin. Therefore one can not apply *Picard's* uniqueness theorem.

1.7 Show that the equation

$$y' = p(t) \cos y + q(t) \sin y,$$

where p, q are continuous functions on $-\infty < t < +\infty$, has a unique solution ϕ on $-\infty < t < +\infty$, satisfying an arbitrary initial condition $\phi(t_0) = y_0$.

Solution.

Consider the IVP

$$\phi'(t) = f(t, \phi(t)), \quad \phi(t_0) = y_0,$$

where

$$f(t, y) = p(t) \cos y + q(t) \sin y,$$

with $p(\cdot), q(\cdot)$ continuous on $-\infty < t < \infty$.

Since $\cos y$ and $\sin y$ are globally *Lipschitz* on $\mathbb{R}$, it follows that $f(t, \cdot)$ is *Lipschitz* in y on each bounded strip

$$D_L = \{(t, y) : |t - t_0| \leq L, y \in \mathbb{R}\},$$

with *Lipschitz* constant $K = K(L)$.

By *Picard–Lindelöf* theorem, there exists a unique solution $\phi(t)$ on $|t - t_0| \leq L$, and moreover ϕ satisfies the estimate

$$|\phi(t)| < |y_0| + \frac{M}{K}(e^{KL} - 1), \quad \forall t \in [t_0 - L, t_0 + L],$$

where

$$M = \max_{(t,y) \in D_L} |f(t, y)|.$$

The above estimate comes from the fact that the *Picard* iterates

$$\phi_0(t) = y_0, \quad \phi_j(t) = y_0 + \int_{t_0}^t f(s, \phi_{j-1}(s)) ds,$$

converge uniformly to ϕ on D_L and $\phi_j(t) = y_0 + \sum_{k=1}^j (\phi_k(t) - \phi_{k-1}(t))$ with the estimates

$$|\phi_j(t) - \phi_{j-1}(t)| \leq \frac{MK^j L^{j+1}}{(j+1)!}, \quad |t - t_0| < L.$$

Since for any $L > 0$ there is a unique bounded solution on D_L , we can extend the solution uniquely on the whole real line $\mathbb{R}$.

Ordinary Differential Equations

ASSIGNMENT 4 – MA 417-2025

TOPIC : BOUNDARY VALUE PROBLEMS.
 02.10.2025

1.1 Find the eigenvalues and eigenfunctions of the following boundary value problems:

- (a) $y''(t) + \lambda y(t) = 0$, on $(0, 1)$, with boundary conditions $y(0) = 0$ and $y(1) + y'(1) = 0$.
 (b) $y''(t) + \lambda y(t) = 0$, on $(0, 2\pi)$, with boundary conditions $y(0) = y(2\pi)$ and $y'(0) = y'(2\pi)$.

Solution. No non-positive eigenvalues for (a) and no negative eigenvalues for (b).

(a) $\lambda_n = k_n^2$, where $k_n > 0$ satisfies $\tan k_n = -k_n$; $y_n(t) = \sin(k_n t)$.

(b) $\lambda_n = n^2$, ($n = 0, 1, 2, \dots$); eigenfunctions: $\begin{cases} y_0(t) = 1, & (n = 0), \\ y_n(t) = \cos(nt), \sin(nt), & (n \geq 1). \end{cases}$

1.2 Show that the boundary value problem $y''(t) + \lambda y(t) = 0$ with $y(0) - y(\pi) = 0$ and $y'(0) + y'(\pi) = 0$ is not self-adjoint and every complex number is eigenvalue of this problem.

Solution. For $L[y] = -y''$,

$$\langle Ly, z \rangle - \langle y, Lz \rangle = [y' \bar{z} - y \bar{z}']_0^\pi.$$

Using $y(\pi) = y(0)$ and $y'(0) = -y'(\pi)$,

$$[y' \bar{z} - y \bar{z}']_0^\pi = 2y'(\pi) \bar{z}(0) \neq 0$$

in general. Hence the boundary form does not vanish, so the operator is not self-adjoint.

Every complex number is an eigenvalue. General solution is given by

$$y(t) = Ae^{ikt} + Be^{-ikt}, \quad k^2 = \lambda.$$

Boundary conditions give

$$\begin{cases} A(1 - e^{i\pi k}) + B(1 - e^{-i\pi k}) = 0, \\ A(1 + e^{i\pi k}) + B(-1 - e^{-i\pi k}) = 0. \end{cases}$$

The determinant of the coefficient matrix is always zero, so nontrivial (A, B) exist for all $k \in \mathbb{C}$. Thus every $\lambda = k^2 \in \mathbb{C}$ is an eigenvalue.

1.3 Consider the Sturm-Liouville differential equation

$$[p(x)y']' - q(x)y + \lambda r(x)y = 0, \quad 0 < x < 1,$$

where the functions $p(x)$, $p'(x)$, $q(x)$, and $r(x)$ are real-valued and continuous on $[0, 1]$. Assume that

$$p(x) > 0, \quad r(x) > 0 \quad \text{for all } x \in [0, 1].$$

The boundary conditions are

$$\alpha_1 y(0) + \alpha_2 y'(0) = 0, \quad \beta_1 y(1) + \beta_2 y'(1) = 0,$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are real constants such that

$$\alpha_1^2 + \alpha_2^2 \neq 0, \quad \beta_1^2 + \beta_2^2 \neq 0.$$

- (a) Check whether the boundary value problem is self-adjoint.
 (b) Show that the eigenvalues λ of the boundary value problem are all real and distinct.

Solution (a). For $L[y] = -(py')' + qy$,

$$\int_0^1 (L[y])\bar{z} - y\overline{L[z]} dx = [p(y'\bar{z} - y\bar{z}')]_0^1.$$

Under the boundary conditions $\alpha_1 y(0) + \alpha_2 y'(0) = 0$, $\beta_1 y(1) + \beta_2 y'(1) = 0$, we have $y'(0) = -(\alpha_1/\alpha_2)y(0)$ (if $\alpha_2 \neq 0$) and similarly at $x = 1$, so the boundary term vanishes. Hence L is **self-adjoint**. Note $\alpha_2 = 0$ case is trivial.

Solution (b). For an eigenfunction y with eigenvalue λ ,

$$(L[y]) + \lambda ry = 0.$$

Multiply by $\bar{y}$, integrate, and subtract the conjugate:

$$(\lambda - \bar{\lambda}) \int_0^1 r|y|^2 dx = 0,$$

so λ is **real**.

If y_1, y_2 correspond to the same λ , their Wronskian $W = y_1 y_2' - y_1' y_2$ satisfies $(pW)' = 0$. From the boundary conditions $W(0) = W(1) = 0 \Rightarrow W \equiv 0$, so y_1, y_2 are linearly dependent.

1.4 Reduce the BVP in self-adjoint BVP form:

(a) Hermite equation: $y''(t) - 2ty'(t) + \lambda y(t) = 0$ with $y(0) = 0$ and $y(1) = 0$.

(b) Euler equation: $t^2 y''(t) + ty'(t) + \lambda y(t) = 0$ with $y(1) = 0$ and $y(e) = 0$.

Solution. A second-order linear ODE $y'' + a(t)y' + b(t)y = 0$ can be rendered self-adjoint by multiplying the equation by the integrating factor $p(t) = e^{\int a(t) dt}$. Then $(py')' + pby = 0$, i.e. in Sturm–Liouville form $-(py')' + qy = \lambda ry$ with appropriate q, r (and $p > 0$ on the interval).

(a) Hermite equation. $y'' - 2ty' + \lambda y = 0$, $y(0) = 0$, $y(1) = 0$. Here $a(t) = -2t \Rightarrow p(t) = e^{\int -2t dt} = e^{-t^2}$. Multiplying gives

$$(e^{-t^2} y')' + \lambda e^{-t^2} y = 0 \iff -(e^{-t^2} y')' + 0 \cdot y = \lambda e^{-t^2} y.$$

Sturm–Liouville data: $p(t) = e^{-t^2} > 0$, $q(t) = 0$, $r(t) = e^{-t^2} > 0$ on $[0, 1]$ with Dirichlet BCs (self-adjoint).

(b) Euler equation. $t^2 y'' + ty' + \lambda y = 0$, $y(1) = 0$, $y(e) = 0$. Divide by t^2 ($t > 0$): $y'' + \frac{1}{t}y' + \frac{\lambda}{t^2}y = 0$, so $a(t) = 1/t \Rightarrow p(t) = e^{\int (1/t) dt} = t$. Multiplying gives

$$(ty')' + \frac{\lambda}{t}y = 0 \iff -(ty')' + 0 \cdot y = \lambda \left(\frac{1}{t}\right)y.$$

Sturm–Liouville data: $p(t) = t > 0$, $q(t) = 0$, $r(t) = 1/t > 0$ on $[1, e]$ with Dirichlet BCs (self-adjoint).

1.5 Consider the boundary value problem $y'' + \lambda y = 0$ in $[0, \pi]$ with the boundary condition

$$m_{11}y(0) + m_{12}y'(0) + n_{11}y(\pi) + n_{12}y'(\pi) = 0$$

and

$$m_{21}y(0) + m_{22}y'(0) + n_{21}y(\pi) + n_{22}y'(\pi) = 0,$$

where m_{ij} and n_{ij} are real constants such that $m_{11}m_{22} - m_{12}m_{21} \neq 0$ or $n_{11}n_{22} - n_{12}n_{21} \neq 0$.

(a) Show that this boundary value problem is self-adjoint if and only if the following condition is satisfied

$$m_{11}m_{22} - m_{12}m_{21} = n_{11}n_{22} - n_{12}n_{21}.$$

(b) Under the same condition, show that the problem has no complex eigenvalues.

Solution. Let

$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}, \quad N = \begin{pmatrix} n_{11} & n_{12} \\ n_{21} & n_{22} \end{pmatrix}, \quad u = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix}, \quad v = \begin{pmatrix} y(\pi) \\ y'(\pi) \end{pmatrix}.$$

The boundary conditions are $Mu + Nv = 0$ and $\det M \neq 0, \det N \neq 0$.

Self-adjointness. The Green boundary form is

$$B[y, z] = [y'\bar{z} - y\bar{z}']_0^\pi = u^T J \bar{u} - v^T J \bar{v}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Using $v = -N^{-1}Mu$,

$$B[y, z] = u^T (J - (N^{-1}M)^T J (N^{-1}M)) \bar{u}.$$

Hence $B[y, z] = 0$ for all y, z iff

$$(N^{-1}M)^T J (N^{-1}M) = J.$$

For any 2×2 real matrix A , $A^T J A = (\det A)J$; applying this gives

$$\det(N^{-1}M) = 1 \iff \det M = \det N.$$

Real eigenvalues. Multiplying $y'' + \lambda y = 0$ by $\bar{y}$ and integrating,

$$[y'\bar{y} - y\bar{y}']_0^\pi + (\lambda - \bar{\lambda}) \int_0^\pi |y|^2 dt = 0.$$

For self-adjoint BCs, the boundary term vanishes, giving

$$(\lambda - \bar{\lambda}) \int_0^\pi |y|^2 dt = 0.$$

Hence, nontrivial solutions require $\lambda = \bar{\lambda}$, i.e. eigenvalues are real.

- 1.6** Find the appropriate Green's function and express the solution of the following boundary value problem as a definite integral:

$$-(y'' + y) = 1; \quad y(0) = 0 \text{ and } y(1) = 1.$$

Solution. Green's function and the solution are given as

$$G(t, s) = \begin{cases} -\frac{\sin t \sin(1-s)}{\sin 1}, & t \leq s, \\ -\frac{\sin s \sin(1-t)}{\sin 1}, & s \leq t, \end{cases} \quad y(t) = \frac{\sin t}{\sin 1} - \int_0^1 G(t, s) ds.$$

- 1.7** Consider the regular Sturm–Liouville boundary value problem:

$$L[y](t) = -\frac{d}{dt} \left(p(t) \frac{dy}{dt} \right) + q(t)y = f(t), \quad t \in [0, 1],$$

where

- $p \in C^1([0, 1])$ with $p(t) > 0$ for all $t \in [0, 1]$,
- $q, f \in C([0, 1])$.

The problem is subject to the boundary conditions:

$$U_1(y) = \alpha_1 y(0) + \alpha_2 y'(0) = 0, \quad U_2(y) = \beta_1 y(1) + \beta_2 y'(1) = 0,$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$ satisfy

$$\alpha_1^2 + \alpha_2^2 \neq 0, \quad \beta_1^2 + \beta_2^2 \neq 0.$$

Assume that $\lambda = 0$ is **not** an eigenvalue of the homogeneous problem

$$L[y] = \lambda y$$

with the same boundary conditions $U_1(y) = 0$ and $U_2(y) = 0$.

(a) Prove that there exists a Green's function $G(t, s)$ such that the solution φ to the inhomogeneous problem can be expressed as

$$\varphi(t) = \int_0^1 G(t, s) f(s) ds.$$

Moreover, show that $G(t, s)$ is given by

$$G(t, s) = \begin{cases} -\frac{y_1(s)y_2(t)}{p(t)W(y_1, y_2)(t)}, & 0 \leq s \leq t \leq 1, \\ -\frac{y_1(t)y_2(s)}{p(s)W(y_1, y_2)(s)}, & 0 \leq t \leq s \leq 1, \end{cases}$$

where

- y_1, y_2 are non-trivial solutions of $L[y] = 0$ satisfying the boundary conditions at $t = 0$ and $t = 1$ respectively,
- $W(y_1, y_2)(t)$ is the Wronskian of y_1 and y_2 .

(b) Show that the Green's function satisfies the jump condition in its partial derivative at $t = s$:

$$\lim_{t \rightarrow s^+} \frac{\partial}{\partial t} G(t, s) - \lim_{t \rightarrow s^-} \frac{\partial}{\partial t} G(t, s) = -\frac{1}{p(s)}.$$

(c) Using the fact that $p(t)W(y_1, y_2)(t)$ is constant on $[0, 1]$, prove that the Green's function G is symmetric; that is,

$$G(t, s) = G(s, t).$$

Solution. Consider

$$L[y](t) := -\frac{d}{dt}(p(t)y'(t)) + q(t)y(t) = f(t), \quad t \in [0, 1],$$

with $p \in C^1$, $p > 0$, $q, f \in C$, and separated boundary conditions $U_1(y) = \alpha_1 y(0) + \alpha_2 y'(0) = 0$, $U_2(y) = \beta_1 y(1) + \beta_2 y'(1) = 0$, where $(\alpha_1, \alpha_2) \neq (0, 0)$, $(\beta_1, \beta_2) \neq (0, 0)$. Assume $\lambda = 0$ is not an eigenvalue of the homogeneous problem $L[y] = \lambda y$ with the same BCs.

(a) **Existence and formula for G .** Let y_1 be a nontrivial solution of $L[y] = 0$ satisfying the left BC $U_1(y_1) = 0$, and y_2 a nontrivial solution satisfying the right BC $U_2(y_2) = 0$. For the Wronskian $W(y_1, y_2) = y_1 y_2' - y_1' y_2$ one checks (using $L[y_j] = 0$) that $(pW)' \equiv 0$, hence $p(t)W(y_1, y_2)(t) \equiv C \neq 0$. Define

$$G(t, s) = \begin{cases} -\frac{y_1(s)y_2(t)}{p(t)W(y_1, y_2)(t)}, & 0 \leq s \leq t \leq 1, \\ -\frac{y_1(t)y_2(s)}{p(s)W(y_1, y_2)(s)}, & 0 \leq t \leq s \leq 1. \end{cases}$$

Then $G(\cdot, s)$ satisfies $L_t G(\cdot, s) = 0$ on $(0, s)$ and $(s, 1)$, the boundary conditions $U_1(G(\cdot, s)) = 0$, $U_2(G(\cdot, s)) = 0$, and is continuous at $t = s$. The unique solution of $L[\phi] = f$ with the given BCs is

$$\phi(t) = \int_0^1 G(t, s) f(s) ds.$$

(b) **Jump of $\partial G / \partial t$ at $t = s$.** Integrate $L_t G(t, s) = \delta(t-s)$ across $t = s$: $\int_{s-\varepsilon}^{s+\varepsilon} (-(pG_t)' + qG) dt = 1$. Letting $\varepsilon \downarrow 0$ gives the jump condition

$$-[p(t)G_t(t, s)]_{t=s^-}^{t=s^+} = 1 \iff \lim_{t \rightarrow s^+} G_t(t, s) - \lim_{t \rightarrow s^-} G_t(t, s) = -\frac{1}{p(s)}.$$

(c) **Symmetry.** Since $p(t)W(y_1, y_2)(t) \equiv C$, we can equivalently write $G(t, s) = -\frac{y_1(\min\{t, s\})y_2(\max\{t, s\})}{C}$, which is manifestly symmetric in t, s . Hence $G(t, s) = G(s, t)$.

Ordinary Differential Equations

ASSIGNMENT 5 – MA 417-2025

TOPIC : LINEAR SYSTEM OF ODES, COMPARISON AND OSCILLATION.
26.09.2025

1.8 Find the exponential operator and the general solution of the system $Y'(t) = AY(t)$ for

(a) $A = \begin{bmatrix} 2 & -2 \\ 1 & 0 \end{bmatrix}$.

(b) $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$.

Solution (a). The characteristic polynomial is $\chi_A(\lambda) = \lambda^2 - 2\lambda + 2 = (\lambda - 1)^2 + 1$, so the eigenvalues are $1 \pm i$. Let $B := A - I$. Then $B^2 = -I$. Hence

$$e^{tA} = e^{t(I+B)} = e^t e^{tB} = e^t (I \cos t + B \sin t) = e^t (I \cos t + (A - I) \sin t).$$

Since $A - I = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$, we obtain

$$e^{tA} = e^t \begin{bmatrix} \cos t + \sin t & -2 \sin t \\ \sin t & \cos t - \sin t \end{bmatrix}.$$

The general solution is $Y(t) = e^{tA}C$ with an arbitrary constant vector $C \in \mathbb{R}^2$.

Solution (b). Here $\chi_A(\lambda) = \lambda(\lambda + 2)$ so the eigenvalues are 0 and -2 . One checks $A^2 = -2A$. Thus

$$e^{tA} = I + \frac{1 - e^{-2t}}{2} A = \frac{1}{2} \begin{bmatrix} 1 + e^{-2t} & 1 - e^{-2t} \\ 1 - e^{-2t} & 1 + e^{-2t} \end{bmatrix}.$$

Equivalently, with eigenvectors $v_0 = (1, 1)^\top$ and $v_{-2} = (1, -1)^\top$, the general solution is

$$Y(t) = c_1 v_0 + c_2 e^{-2t} v_{-2}, \quad c_1, c_2 \in \mathbb{R}.$$

1.9 Consider the equation $ay'' + by' + cy = 0$, where $a(\neq 0)$, b and c are constants. Transform the given equation into system ($Y' = AY$) and then find the eigenvalues of the matrix A . What is the relation between roots of the characteristic equation $ar^2 + br + c = 0$ and eigenvalues of A ?

Solution. Set $x_1 = y$, $x_2 = y'$. Then

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =: A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

The characteristic polynomial of A is

$$\chi_A(\lambda) = \det(\lambda I - A) = \lambda^2 + \frac{b}{a} \lambda + \frac{c}{a} = \frac{1}{a} (a\lambda^2 + b\lambda + c).$$

Hence the eigenvalues of A are exactly the roots of the scalar characteristic equation

$$ar^2 + br + c = 0,$$

so the eigenvalues of A coincide with the ODE characteristic roots.

1.10 Let $A_{n \times n}$ be a real matrix. Show that $\det(e^{tA}) = e^{t(\text{trace } A)}$. Conclude that e^{tA} is a nonsingular matrix. Show that for each $t \in \mathbb{R}^n$, $\lim_{Y \rightarrow \xi} e^{tA}Y = e^{tA}\xi$ (i.e. $e^{tA} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous). Also show that for each $t \in \mathbb{R}$, the mapping $\xi \mapsto e^{tA}\xi$, from $\mathbb{R}^n$ to $\mathbb{R}^n$, is continuously differentiable.

Solution. Let $X(t) := e^{tA}$. By Jacobi's formula,

$$\frac{d}{dt} \det X(t) = \det X(t) \operatorname{tr}(X(t)^{-1} X'(t)).$$

Since $X'(t) = AX(t)$ and $X(t)^{-1} X'(t) = A$, we get

$$\frac{d}{dt} \det(e^{tA}) = \operatorname{tr}(A) \det(e^{tA}), \quad \det(e^{0 \cdot A}) = 1.$$

Solving this scalar ODE yields

$$\det(e^{tA}) = e^{t \operatorname{tr} A}.$$

Therefore $\det(e^{tA}) \neq 0$ for all t , so e^{tA} is invertible (indeed $(e^{tA})^{-1} = e^{-tA}$).

Fix t . The map $Y \mapsto e^{tA}Y$ is linear, hence continuous. Explicitly, for any operator norm,

$$\|e^{tA}Y - e^{tA}\xi\| \leq \|e^{tA}\| \|Y - \xi\| \rightarrow 0 \quad (Y \rightarrow \xi).$$

Finally, for fixed t , $\xi \mapsto e^{tA}\xi$ is linear with constant derivative $D(e^{tA} \cdot)_\xi[h] = e^{tA}h$; thus it is C^1 (indeed C^∞) in ξ .

- 1.11** Let $\phi(t) = \begin{bmatrix} \sin t & \cos t \\ -\cos t & \sin t \end{bmatrix}$ be a fundamental matrix for the system $Y'(t) = AY(t)$. Find constant matrix $A_{n \times n}$ and compute e^{tA} .

Solution. Any fundamental matrix satisfies $\phi'(t) = A\phi(t)$. Hence

$$A = \phi'(t)\phi(t)^{-1} \quad (\text{independent of } t); \text{ in particular at } t = 0, \quad A = \phi'(0)\phi(0)^{-1}.$$

Compute $\phi'(t) = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$, so $\phi'(0) = I$ and $\phi(0) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ with inverse $\phi(0)^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Therefore

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then $e^{tA} = \phi(t)\phi(0)^{-1}$, i.e.

$$e^{tA} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix},$$

the standard rotation matrix.

- 1.12** Consider $Y'(t) = AY(t)$, where A is $n \times n$ matrix. Let $\phi(\cdot)$ be a fundamental matrix of the system. Show that any matrix $\psi(\cdot) \in M_{n \times n}$ is also a fundamental matrix if and only if there exists a nonsingular matrix $C \in M_{n \times n}$ such that $\psi(t) = \phi(t)C$, for all $t \in \mathbb{R}$.

Solution. ($\Rightarrow$) Suppose ψ is another fundamental matrix. Define $C(t) := \phi(t)^{-1}\psi(t)$. Differentiate, using $\phi' = A\phi$ and $\psi' = A\psi$:

$$C'(t) = \frac{d}{dt}(\phi^{-1}\psi) = -\phi^{-1}\phi'\phi^{-1}\psi + \phi^{-1}\psi' = -\phi^{-1}A\psi + \phi^{-1}A\psi = 0.$$

Hence $C(t) \equiv C$ is constant; since ψ is fundamental, C is invertible, and $\psi(t) = \phi(t)C$.

($\Leftarrow$) If $\psi(t) = \phi(t)C$ with C constant and invertible, then $\psi'(t) = \phi'(t)C = A\phi(t)C = A\psi(t)$, and $\psi(t)$ is invertible whenever $\phi(t)$ is. Thus ψ is a fundamental matrix.

- 1.13** (a) Let $f(t) = t^a e^{-bt}$, for some positive constants a and b , and let c be a positive number smaller than b . Show that we can find a positive constant K such that $|f(t)| \leq K e^{-ct}$, $0 \leq t < \infty$.

- (b) Let $A = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$, where $\lambda < 0$. Show that there exist positive constants K and α such that $\|e^{tA}X_0\| \leq K e^{-\alpha t} \|X_0\|$, for all $X_0 \in \mathbb{R}^2$. (Hint: Use the expression for e^{tA} and part (a)).

Solution.

(a) Write

$$t^a e^{-bt} = e^{-ct} \underbrace{t^a e^{-(b-c)t}}_{=:g(t)}.$$

For $\alpha := b - c > 0$, $g(t) = t^a e^{-\alpha t}$ achieves its maximum at $t_* = a/\alpha$ with value $g(t_*) = (a/\alpha)^a e^{-a}$. Taking $K := (a/\alpha)^a e^{-a}$ yields

$$t^a e^{-bt} \leq K e^{-ct} \quad \text{for all } t \geq 0.$$

(b) Decompose $A = \lambda I + N$ with $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $N^2 = 0$. Then

$$e^{tA} = e^{\lambda t} e^{tN} = e^{\lambda t} (I + tN) = e^{\lambda t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

In the induced ∞ -norm, $\left\| \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \right\|_\infty = 1 + t$. Fix $\alpha \in (0, -\lambda)$ and let $\beta := -\lambda - \alpha > 0$.

For $t \geq 0$,

$$\|e^{tA}\|_\infty \leq e^{\lambda t} (1 + t) = e^{-\alpha t} (e^{-\beta t} + t e^{-\beta t}) \leq e^{-\alpha t} \left(1 + \sup_{t \geq 0} t e^{-\beta t} \right) \leq e^{-\alpha t} \left(1 + \frac{1}{e\beta} \right).$$

Thus, with $K := 1 + \frac{1}{e(-\lambda-\alpha)}$,

$$\|e^{tA} X_0\|_\infty \leq K e^{-\alpha t} \|X_0\|_\infty \quad (t \geq 0).$$

By equivalence of norms on $\mathbb{R}^2$, the same estimate holds for any fixed vector norm after adjusting K .

1.14 Let $A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & a \end{bmatrix}$. If $a < 0$, show that for each initial value $X_0 \in \mathbb{R}^3$, there is a positive constant M such that the solution of the IVP satisfies $\|e^{tA} X_0\| \leq M \|X_0\|$ for all $t \geq 0$.

Solution. The matrix is block diagonal with a planar rotation generator and a scalar a . Hence

$$e^{tA} = \begin{bmatrix} \cos t & \sin t & 0 \\ -\sin t & \cos t & 0 \\ 0 & 0 & e^{at} \end{bmatrix} = \text{diag}(R(t), e^{at}), \quad R(t) \in SO(2).$$

With the Euclidean norm, $\|R(t)\|_2 = 1$ for all t , and $0 < e^{at} \leq 1$ for $t \geq 0$ since $a < 0$. Therefore $\|e^{tA}\|_2 = \max\{1, e^{at}\} \leq 1$ for all $t \geq 0$. Consequently,

$$\|e^{tA} X_0\|_2 \leq \|X_0\|_2 \quad \text{for all } t \geq 0.$$

If a different norm is used, $\sup_{t \geq 0} \|R(t)\|$ is still finite (e.g. $\leq \sqrt{2}$ for $\|\cdot\|_1$ or $\|\cdot\|_\infty$), so one may take $M = \max\{\sup_{t \geq 0} \|R(t)\|, 1\}$.

Comparison and Oscillation.

1.15 Show that for $0 \leq a < \infty$, any nontrivial solution of the Hermite equation

$$u''(t) - 2t u'(t) + 2a u(t) = 0, \quad t \in \mathbb{R},$$

has at most finitely many zeros.

Hint: Consider the function $v(t) := e^{-t^2/2} u(t)$, and determine the differential equation satisfied by $v(t)$.

Solution. Let $v(t) = e^{-t^2/2}u(t)$. Then $u = e^{t^2/2}v$, and a direct computation gives

$$v''(t) + ((2a+1) - t^2)v(t) = 0. \quad (1)$$

Fix $c > 0$ and choose $R = \sqrt{2a+1+c}$. For $|t| \geq R$, we have $q_v(t) = (2a+1) - t^2 \leq -c$. Compare with $z'' - cz = 0$, whose nontrivial solutions are non-oscillatory (at most one zero on each half-line). Since $q_v(t) \leq -c$ on $(R, \infty) \cup (-\infty, -R)$, by the *Sturm comparison theorem*, any solution of (1) has at most one zero on each of these tails.

On the bounded interval $[-R, R]$, the coefficient $q_v(t)$ is analytic, so v is analytic and can have only finitely many zeros there. Therefore, v (and hence $u = e^{t^2/2}v$) has at most finitely many zeros.

1.16 Let $q : (0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that

$$q(t) \geq \frac{c}{t^2} \quad \text{for some constant } c > \frac{1}{4}.$$

Show that any nontrivial solution u of the differential equation

$$u''(t) + q(t)u(t) = 0$$

has infinitely many zeros in $(0, \infty)$.

Hint. Let u be a solution of $u''(t) + \frac{c}{t^2}u(t) = 0$. Consider the equation satisfied by $v(t) := e^{-t/2}u(e^t)$.

Solution. Let $v(t) = e^{-t/2}u(e^t)$ with $x = e^t > 0$. Then the equation $u''(x) + \frac{c}{x^2}u(x) = 0$ transforms into

$$v''(t) + (c - \frac{1}{4})v(t) = 0.$$

Since $c > \frac{1}{4}$, every nontrivial solution is oscillatory:

$$v(t) = A \cos(\omega t) + B \sin(\omega t), \quad \omega = \sqrt{c - \frac{1}{4}} > 0,$$

and hence v has infinitely many zeros in $\mathbb{R}$. Because $u(e^t) = e^{t/2}v(t)$, zeros of v correspond bijectively to zeros of u on $(0, \infty)$; thus any nontrivial solution of

$$u''(x) + \frac{c}{x^2}u(x) = 0$$

has infinitely many zeros in $(0, \infty)$.

Now, if $q(t) \geq \frac{c}{t^2}$ with $c > \frac{1}{4}$, let u solve $u'' + q(t)u = 0$ and w solve $w'' + \frac{c}{t^2}w = 0$. By the *Sturm comparison theorem*, between any two consecutive zeros of w , the function u must have at least one zero. Since w has infinitely many zeros, u must also have infinitely many zeros on $(0, \infty)$.

1.17 Let $q : [a, b] \rightarrow \mathbb{R}$ be a continuous function and let $u, v \in C^1([a, b])$ be twice differentiable on (a, b) . Suppose that

$$u(0) = v(0) = 0, \quad u'(0) = v'(0) > 0, \quad u(t) > 0 \text{ for } t \in (a, b),$$

and that

$$u''(t) + q(t)u(t) = 0, \quad t \in [a, b],$$

$$v''(t) + q(t)v(t) \geq 0, \quad t \in [a, b].$$

Show that

$$v(t) \geq u(t) \quad \text{for all } t \in [a, b].$$

Solution. Let $w = v - u$. Then

$$w'' + qw = (v'' + qv) - (u'' + qu) \geq 0, \quad w(0) = w'(0) = 0.$$

Define

$$\Phi(t) := w'(t)u(t) - w(t)u'(t).$$

Differentiating and using $u'' + qu = 0$ gives

$$\Phi'(t) = w''(t)u(t) - w(t)u''(t) = (w''(t) + q(t)w(t))u(t) \geq 0,$$

since $u(t) > 0$ on (a, b) . Thus Φ is non-decreasing, and $\Phi(0) = 0$, so $\Phi(t) \geq 0$ for all t .

Then

$$\left(\frac{w}{u}\right)' = \frac{\Phi}{u^2} \geq 0,$$

so w/u is non-decreasing on (a, b) . As $w(0) = 0$, it follows that $w/u \geq 0$, hence $w(t) \geq 0$ for all $t \in [a, b]$. Therefore, $v(t) \geq u(t)$ for all $t \in [a, b]$.